Jeongrak Son

Postdoctoral Researcher – Dahlem Center for Complex Quantum Systems – Freie Universität Berlin

@ jeongrak.son@gmail.com   Berlin, Germany   Personal Website

Education

Ph.D. in Physics
Nanyang Technological University (NTU)
Aug. 2021 – Nov. 2025   Singapore, Singapore

Thesis: Quantum qomrades: catalysts in resource theories and memories in dynamic programming **[Link]**
Adviser: Nelly H. Y. Ng

B.Sc. in Physics **Summa Cum Laude**
Seoul National University (SNU)
Mar. 2015 – Feb. 2021   Seoul, Korea

Military service (Rep. of Korea Air Force): Mar. 2018 – Feb. 2020

Exchange Student
Department of Physics and Astronomy - University of Manchester
Sep. 2016 – Feb. 2017   Manchester, United Kingdom

Languages

Korean ●●●●●

English ●●●●●

French ●○○○○

Hobby

Passionate cinephile
Selected film reviews:
- In Perfect Days (2023), It's Okay to Cry (Featured in Asian Film Archive's monthly newsletter)
- Hong Sang-Soo before and after Kim Min-Hee (Proof version; see Exposure Print (Issue 1) for the final version)

Check my other reviews here

Research Interests

- Quantum algorithms: quantum data access models, quantum recursions
- Catalysis in quantum information: catalysis in circuit compilation, state-oblivious catalysis
- Resource theories: compositional structures in resource theories

Work Experiences

Postdoctral Researcher
with Jens Eisert
May. 2026 –   DCCQS, Freie Universität Berlin

Project Officer / Research Fellow
with Nelly H. Y. Ng
May. 2025 – Jan. 2026 / Feb. 2026 – Apr. 2026   SPMS, Nanyang Technological University

IBS Student Trainee / Research Assistant
with Juzar Thingna and Peter Talkner
Jul.2020 – Aug. 2020 / Jan. 2021 – May. 2021   PCS, Institute for Basic Science

Preprints

1. Y. Zeng*, **J. Son***, M. Gu, and X. Yuan, Theory of Quantum Imaginary-Time Mpemba Effect, arXiv:2604.17412 (2026). **[arXiv]** (*: co-first authors)
2. R. Ganardi, **J. Son**, J. Czartowski, S. H. Lie, and N. H. Y. Ng, Manipulating heterogeneous quantum resources over a network, arXiv:2602.17803 (2026). **[arXiv]**

3. X. Hu, L. Lautenbacher, G. Spaventa, M. B. Plenio, N. H. Y. Ng, and **J. Son**, Gaussian time-translation covariant operations: structure, implementation, and thermodynamics, accepted in Phys. Rev. Lett. **[arXiv]**
   1/2 of Quantum Thermodynamics 2025 invited talk (speaker: Xueyuan Hu); selected as a Quantum Resources 2026 talk (speaker: Giovanni Spaventa); selected as an AQIS 2026 long talk (speaker: Xueyuan Hu)
4. Y. Suzuki, M. Gluza, **J. Son**, B. H. Tiang, N. H. Y. Ng, and Z. Holmes, Grover's algorithm is an approximation of imaginary-time evolution, arXiv:2507.15065 (2025). **[arXiv]**
   selected as an IPS meeting 2025 talk (speaker: Bi Hong Tiang); selected as a QCTiP 2026 talk (speaker: Yudai Suzuki)

Published Works

1. M. Robbiati*, E. Pedicillo*, A. Pasquale*, L. Xiaoyue*, O. Kiss*, A. Wright, R. M. S. Farias, K. U. Giang, **J. Son**, J. Knörzer, S. T. Goh, J. Y. Khoo, N. H. Y. Ng, Z. Holmes, S. Carrazza, M. Gluza, Double-bracket quantum algorithms for high-fidelity ground-state preparation, Phys. Rev. Research **8**, 033078 (2026). **[Link]** (*: co-first authors)
2. S. H. Lie, **J. Son**, P. Boes, N. H. Y. Ng, and H. Wilming, Thermal Operations from Informational Equilibrium, Phys. Rev. Lett. **137**, 030403 (2026). **[Link (open access)]**
   Kyoto Workshop of Quantum Thermodynamics and Stochastic Thermodynamics 2025 invited talk (speaker: Nelly Ng); selected as a Quantum Resources 2026 talk (speaker: Seok Hyung Lie)
3. **J. Son**, R. Ganardi, S. Minagawa, F. Buscemi, S. H. Lie, and N. H. Y. Ng, Catalytic Channels Are the Only Noise-Robust Catalytic Processes, Phys. Rev. Lett. **136**, 050202 (2026). **[Link] [arXiv]**
   Quantum Resources 2025 invited talk (speaker: Nelly Ng); selected as an AQIS2025 talk (speaker: Seok Hyung Lie); selected as an IPS meeting 2025 talk; media coverage (in Korean) [1] [2]; CQT highlight article
4. M. Gluza, **J. Son**, B. H. Tiang, R. Zander, R. Seidel, Y. Suzuki, Z. Holmes, and N. H. Y. Ng, Double-Bracket Quantum Algorithms for Quantum Imaginary-Time Evolution, Phys. Rev. Lett. **136**, 020601 (2026). **[Link] [arXiv]**
   selected as a QTD2025 talk; selected as an IPS meeting 2025 invited talk; CQT highlight article
5. Y. Suzuki, B. H. Tiang, **J. Son**, N. H. Y. Ng, Z. Holmes, and M. Gluza, Double-bracket algorithm for quantum signal processing without post-selection, Quantum **9**, 1954 (2025). **[Link]**
   selected as an IPS meeting 2025 talk (speaker: Marek Gluza)
6. **J. Son**, M. Gluza, R. Takagi, and N. H. Y. Ng, Quantum Dynamic Programming, Phys. Rev. Lett. **134**, 180602 (2025). **[Link] [arXiv]**
   honourable mentions for Top quantum algorithms papers in Spring 2024 by PennyLane (Xanadu); selected as an IPS meeting 2024 talk; CQT highlight article
7. **J. Son** and N. H. Y. Ng, A hierarchy of thermal processes collapses under catalysis, Quantum Sci. Technol. **10**, 015011 (2024). **[Link] [arXiv]**
   selected as 1/2 of AQIS2023 talk; selected as a Quantum Resources 2023 talk; selected as a Beyond IID 2024 talk
8. A. de Oliveira Junior*, **J. Son***, J. Czartowski, and N. H. Y. Ng, Entanglement generation from athermality, Phys. Rev. Research **6**, 033236 (2024). **[Link]** (*: co-first authors)
   selected as an IPS meeting 2024 talk; selected as a Quantum Resources 2025 talk
9. **J. Son** and N. H. Y. Ng, Catalysis in action via elementary thermal operations, New J. Phys. **26**, 033029 (2024). **[Link]**
   selected as a Quantum Resources 2022 talk; selected as 1/2 of AQIS2023 talk
10. **J. Son**, P. Talkner, and J. Thingna, Charging quantum batteries via Otto machines: Influence of monitoring, Phys. Rev. A **106**, 052202 (2022). **[Link] [arXiv]**
    selected as 1/2 of QTD2022 talk (speaker: Juzar Thingna); selected as a ICE-7 lightning talk
11. **J. Son**, P. Talkner, and J. Thingna, Monitoring quantum Otto engines, PRX-Quantum **2**, 040328 (2021). **[Link]**
    selected as 1/2 of QTD2022 talk (speaker: Juzar Thingna); media coverage [1] [2]

Talks and Seminars

7 Conference Talks and 16 Seminar Talks

- Conference Talks: **Quantum resources**: from mathematical foundations to operational characterisation (Dec. 2022), **AQIS 2023** (Aug. 2023), **Quantum resources 2023** (Dec. 2023) **(video)**, **Beyond IID 2024** (Jul. 2024), **Quantum resources 2025** (Mar. 2025), **Quantum Thermodynamics 2025** (Jul. 2025), **IPS meeting 2025 [invited]** (Sep. 2025)
- Seminar Talks: **IBS PCS Seminar** (Jun. 2022) **(video)**, **Majulab Seminar** (Dec. 2022), **Chaos and Quantum Info Seminar** [Jagiellonian U.] (Feb. 2023), **CQT Seminar** (May 2023), **QST Seminar** [Seoul Natl. U.] (Jul. 2023), **Informal Statistical Physics Seminar** [U. Maryland] (Aug. 2024) **(abstract)**, **Q.InC Seminar** [A*STAR] (Sep. 2024) **(abstract)**, **KIAS Seminar** [KIAS] (Oct. 2024) **(abstract)**, **AG Eisert group meeting** [FU Berlin] (Jan. 2025) **(video)**, **Institut für Theoretische Physik group meeting** [Uni Ulm] (Jan. 2025), **bigQ - Center for Macroscopic Quantum States** [DTU] (Jan. 2025), **Q-DNA group** [Yonsei U.] (Dec. 2025), **CQT Lunch Seminar** (Jan. 2026), **Department of Quantum Information** [Yonsei U.] (Apr. 2026), **QST Seminar** [Seoul Natl. U.] (Apr. 2026), **KIAS Seminar** [KIAS] (Apr. 2026) **(abstract)**

1 Lightning Talk and 3 Short Talks (<20 mins)

- Lightning Talk: **ICE-7** Quantum Information and Quantum Technologies Conference (May 2022)
- Short Talks: **IPS meeting 2024** [2 contributed talks] (Oct. 2024), **IPS meeting 2025** (Sep. 2024)

Peer Review Contributions

- IOP Outstanding Reviewer Awards 2025 for Quantum Science and Technology **(credential)**; IOP Trusted Reviewer **(credential)**
- Referee for **Phys. Rev. X**, **Phys. Rev. Lett.**, **Quantum**, **Quantum Sci. Technol.**, **Phys. Rev. A**, **Phys. Rev. E**, **J. Math. Phys.**, and **Phys. Scr.**
- Sub-reviewer for **TQC 2022**, **QIP2023**, **TQC 2025**, and **QCTiP 2026**

Academic Visits

- Yonsei University, Korea [host: **Kyunghyun Baek**] (Apr. 2026)
- University of Tokyo, Japan [host: **Ryuji Takagi**] (Mar. 2026)
- Yonsei University, Korea [host: **Daniel K. Park**] (Dec. 2025)
- Ulsan National Institute of Science and Technology (UNIST), Korea [host: **Seok Hyung Lie**] (Nov. 2025)
- Danmarks Tekniske Universitet (DTU), Denmark [host: **Jonatan Bohr Brask**] (Jan. 2025)
- Universität Ulm, Germany [host: **Martin Plenio**] (Jan. 2025)
- Freie Universität Berlin (FU Berlin), Germany [host: **Jens Eisert** and **Nathan Walk**] (Jan. 2025)
- Korea Institute for Advanced Study (KIAS), Korea [host: **Hyukjoon Kwon**] (Oct. 2024, Mar. 2025, and Apr. 2026)
- University of Maryland, Baltimore County, USA [host: **Sebastian Deffner**] (Aug. 2024)
- University of Maryland, College Park, USA [host: **Nicole Yunger Halpern**] (Aug. 2024)
- Nagoya Universtiy, Japan [host: **Francesco Buscemi**] (Jul. 2024)
- Jagiellonian University, Poland [host: **Kamil Korzekwa**] (Feb. 2023)
- PCS, Institute for Basic Science, Korea [host: **Dario Rosa**] (Jun. 2022)

Teaching and Services

Secretary (2023) and Treasurer (2024)

Quantum Young Researchers Association (QYRA), Singapore

2023 – 2024

- Lead organiser of QYRA X Infocom Media Development Authority (IMDA) event ***Careers in Quantum Communications***
- Organiser of ***Quantum Energy Initiative (QEI) workshop 2023***
- Topical team member (w/ Masahito Ueda, Gentaro Watanabe, and Ariane Soret) representing one of the five workshop topics: "Fundamental thermodynamics of information" in the ***Quantum Energy Initiative (QEI) workshop 2023***—moderated discussions throughout the workshop and presented the outcomes of the team's deliberations
- Facilitator for the Townhall event ***Building Singapore's Quantum Future Together****: A Multi-Stakeholder Townhall on the National Quantum Strategy and Entrepreneurship*
- Organiser (logistics) of QYRA's End-of-Year event ***What's next for early-career Qontributors in Singapore?***

Scientific Adviser for Korean Translation of the book series "for babies" by Chris Ferrie

CHAEKSESANG, Korea

2023–2023

Advised translations for 12 books: Quantum Physics, Quantum Information, Quantum Entanglement, Quantum Computing, Optical Physics, Statistical Physics, Electromagnetism, General Relativity, Newtonian Physics, Nuclear Physics, Astrophysics, and Rocket Science

SINGA Ambassador

Singapore International Graduate Award (SINGA), Singapore

2023 – 2023

Teaching Experiences

- Tutorial for PH1107: Relativity and Quantum Physics, Nanyang Technological University, Singapore, AY22/23, 23/24, and 24/25
- Tutorial for International Students, Physics 1, Seoul National University, Korea, AY20
- Organizer/Tutor for Summer Science Camp, Korea Student Aid Foundation and Gyeongsang Girl's High School, Korea, Jul. 2015

Awards and Honours

- **Singapore International Graduate Award (SINGA)** [The Agency for Science, Technology and Research (A*STAR)] (2021 – 2025)
- **GE Foundation Scholar-Leaders Program (GEFSLP) Scholarship** [Fulbright Korea] (2016 – 2021)
- **Presidential Science Scholarship** [Korea Student Aid Foundation] (2015 – 2021)
- **OIA Outgoing Exchange Student Scholarship** [Office of International Affairs (OIA), Seoul National University] (2016)
- **Dean's List** [College of Natural Sciences, Seoul National University] (Autumn 2015, Spring 2020)